

RAMCO INSTITUTE OF TECHNOLOGY
RAJAPALAYAM

ACADEMIC YEAR 2024-2025

DEPT. OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Year and Semester: II/IV

Subject code and Title: CS3591-COMPUTER NETWORKS

Faculty: Dr.M. Kaliappan, Professor & Head

Date: 10.03.2024

Activity name: LAN Explorer with Poster presentation

Procedure for the Activity

1. Divide Students into Teams

- Form small groups (3–5 students each).
- Assign each group a section of the lab or a set of network components.

2. Observation & Data Collection

- Identify and record the number of systems, switches, routers, and access points.
- Use ipconfig, ping, or tracert commands to gather IP, MAC addresses, and hostnames of assigned machines.
- Note cable types, bandwidth, DHCP server, and topology layout.

3. Network Mapping

- Draw a physical + logical network diagram showing connections between switches, routers, servers, and PCs.
- Label each device with its IP, MAC, and role (file server, database server, etc.).

4. Problem-Solving Scenarios

- What if 20 more PCs are added?
- How would you divide the network into 4 subnets?
- Suggest improvements (Wi-Fi upgrade, redundancy, etc.).

5. Presentation & Peer Learning

- Each group presents their findings in poster. Poster was uploaded in LMS
- Other groups ask questions and compare their network section.

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ANS - IP address of the router/gateway: 192.168.1.1

12. Draw the Network Topology (show how machines and servers are connected using connectivity devices)

NETWORK Topology

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graph TD
    Internet[Internet] --- Router[Router]
    Router --- S1[Switch]
    Router --- S2[Switch]
    Router --- S3[Switch]
    S1 --- PCs[PCs]
    S2 --- Servers[Servers]
    S3 --- WiFi[Wi-Fi AP]
  
```

13. If 20 more machines have to be added to the network, what changes must be made to the network?

ANS - **Changes required:** Add an additional switch to handle the extra machines.
IP configuration: Ensure the DHCP server has enough IP addresses in the pool or configure static IPs if needed.

14. If the network is to be divided into four subnetworks having 50 machines each, give a plan to do so. What additional devices will be needed? Give the IP address of each subnetwork and the address ranges for hosts in each subnetwork.

Subnet	Network Address	Range of Host IPs	Broadcast Address
Subnet 1	192.168.1.78/26	192.168.1.1 - 192.168.1.62	192.168.1.63
Subnet 2	192.168.1.54/26	192.168.1.65 - 192.168.1.126	192.168.1.127

Submitted: May 28, 2024 at 1:39pm

Word Count: 388 words

Submitted Files: (click to load)
[m ans LAN.docx](#)

Assessment
 Grade out of 10
 9

Assignment Comments

Paragraph | B I U A | 0 words

Network components
 m - C32014 - Computer Networks(2024-2025) - B SEC.

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COMPUTER NETWORK ASSIGNMENTS - FILLED

Activity: Poster presentation

Study of LAN environment

1. Total Number of computers in your lab: 25

2. Find details of any 5 computers:

Hostname	MAC Address	IP Address	LAN Speed	Default Mask
Lab-PC01	00:1A:2B:3C:4D:01	192.168.0.101	100 Mbps	255.255.255.0
Lab-PC02	00:1A:2B:3C:4D:02	192.168.0.102	100 Mbps	255.255.255.0
Lab-PC03	00:1A:2B:3C:4D:03	192.168.0.103	100 Mbps	255.255.255.0
Lab-PC04	00:1A:2B:3C:4D:04	192.168.0.104	100 Mbps	255.255.255.0
Lab-PC05	00:1A:2B:3C:4D:05	192.168.0.105	100 Mbps	255.255.255.0

3. Are the IP addresses assigned to the machines statically or dynamically? Dynamically

4. Does the network have a DHCP server? Yes

5. If yes, what is the address of the server? 192.168.0.1

6. How many servers are configured? : 3

Details of servers :

IP Address	MAC Address	Purpose
192.168.0.1	00:1A:2B:00:00:01	DHCP & Gateway
192.168.0.2	00:1A:2B:00:00:02	File Server
192.168.0.3	00:1A:2B:00:00:03	Database Server

Submitted: May 21 at 5:21pm

Word Count: 93 words

Submitted Files: (click to load)
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Assessment
 Grade out of 10
 10

Assignment Comments

Paragraph | B I U A | 0 words

Reflective Critique

Strengths of the Activity:

- Encourages experiential learning: students don't just read about networks; they discover real setups.
- Promotes teamwork and critical thinking (e.g., subnetting, capacity planning).
- Connects theory (IP addressing, DHCP, topology) to practical lab environment.
- Builds confidence in using network commands and tools.

Challenges Faced by Students:

- Some may struggle with technical commands (ipconfig, tracert).

- Network details (like DHCP scope or switch configuration) may not be fully accessible to students.
- Limited time may prevent deep exploration of larger networks.

Improvements / Future Enhancements:

- Provide students with guided worksheets (tables for IPs, MACs, switch details).
- Introduce simulation tools (Cisco Packet Tracer / GNS3) so students can redesign the network virtually.
- Encourage reflection journals where students write what they learned and difficulties faced.
- In advanced versions, ask students to troubleshoot a pre-created network problem


10/3/24
Faculty Incharge


10/3/24
HOD/AD

COMPUTER NETWORK ASSIGNMENTS

Assignment : Poster presentation

Study of LAN environment

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Find out information about the network in your lab and fill in details below:

Find out information about the network in your lab and fill in details below:

1. Total Number of computers in your lab: 60

2. Find details of any 5 computers:

	MAC address	IP address	LAN speed	Default mask	hostname
1	00-BE-43-ED-29-1F	172.16.71.44	1000	255.255.240.0	RIT-AIDS-44
2	00-BE-43-ED-3E-96	172.16.71.43	1000	255.255.240.0	RIT-AIDS-43
3	00-BE-43-ED-3F-9F	172.16.71.50	1000	255.255.240.0	RIT-AIDS-50

4	00-BE-43-ED-2C-FC	172.16.71.47	1000	255.255.240.0	RIT-AIDS-47
5	00-BE-43-ED-37-48	172.16.71.48	1000	255.255.240.0	RIT-AIDS-48

3. Are the IP addresses assigned to the machines statically or dynamically?

Dynamically (via DHCP)

4. Does the network have a DHCP server?

Yes , the network has a DHCP server enabled in it.

5. If yes, what is the address of the server ?

The address of the DHCP server is 172.16.0.86

6.

How many servers are configured? : Details of servers :

	IP address	MAC address	Purpose
1	172.16.71.44	00:1A:2B:3C:4D:63	File Server
2	172.16.71.43	00:1A:2B:3C:4D:64	Web Server
3	172.16.71.50	00:1A:2B:3C:4D:65	Database Server

7. Cables

a. Type : **CAT 6**

b. Is it coaxial / twisted pair or fiber optic cable ?

Twisted pair

c. Cable bandwidth : **1 Gbps**

d.
Maximum cablelength limit
:

100 meters for Cat6

e. Connector used : **RJ45**

8. Switches:

No	Company Name	MAC address	No. of ports	Managed / Unmanaged	IP's of Machines connected to the switch
1	Cisco	00:1A:2B:3C:4D:66	24	Managed	172.16.71.44
2	TP-Link	00:1A:2B:3C:4D:67	8	Unmanaged	172.16.71.43
3	Netgear	00:1A:2B:3C:4D:68	16	Managed	172.16.71.50
4	D-Link	00:1A:2B:3C:4D:69	24	Unmanaged	172.16.71.47
5	Linksys	00:1A:2B:3C:4D:70	8	Managed	172.16.71.48

9. Routers:

No	Company Name	No. / Types of ports	Port speed	IP address	No
1	Cisco	4 Ethernet, 1 WAN	1 Gbps	172.16.71.44	1
2	D-Link	4 Ethernet, 1 WAN	1 Gbps	172.16.71.43	2
3	TP-Link	4 Ethernet, 1 WAN	1 Gbps	172.16.71.50	3

10. Is there wi-fi capability in the LAN?

If yes,

i. What is the Wi-fi access point address?

Wi-Fi access point address: 192.168.1.100

ii. How many devices / IP addresses does it support?

Number of devices / IP addresses it supports: 50 devices

iii. What is the bandwidth?
h?

Bandwidth : 300 Mbps

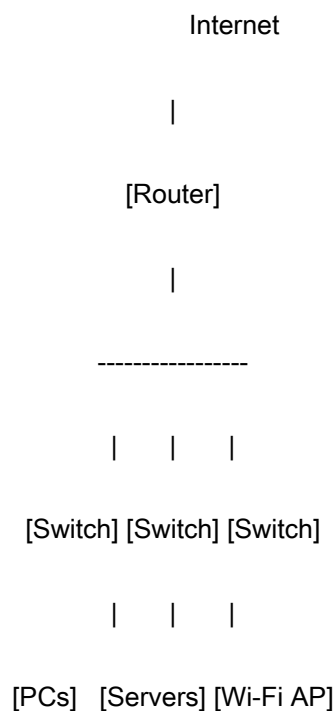
11. Is there internet access in the lab? If not, what changes to the hardware / software must be made ?
If yes, what is the IP address of the router / gateway ?

Yes, it has internet access

IP address of the router/gateway: 192.168.1.1

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Network Topology :



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Subnet 3	172.16.71.128/26	: 172.16.71.129 - 172.16.71.190	172.16.71.191
Subnet 4	172.16.71.192/26	172.16.71.193 - 172.16.71.254	172.16.71.255